## ① Qiskit program

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)          fault
7 circuit.cx(0,1)
8 circuit.measure_all()
```

*lines 10–16: simulate, print*

*Expected* $(|01\rangle+|10\rangle)/\sqrt{2}$*; line 6 should apply X.*

## ② srcML of line 7

```
<expr_stmt pos:start="7:1"[d]>
 <expr><call>[a]
  <name><name>circuit</name>
   <operator>.</operator><name>cx[b]</name></name>
  <argument_list>(<argument><expr>
   <literal type="number">0[c]</literal>
   </expr></argument>,<argument><expr>
   <literal type="number">1</literal>
   </expr></argument>)</argument_list>
 </call></expr></expr_stmt>
```

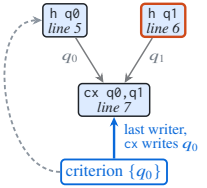
**a** `call` → one operation **b** `cx` → gate
**c** `0`, `1` → $q_0$ control, $q_1$ target
**d** position → line 7 the QDG writes both

## ③ Trace

| line | gate | qubits |
|---|---|---|
| 5 | h | $q_0$ |
| 6 | h | $q_1$ |
| 7 | cx | $q_0, q_1$ |
| 8 | measure | $q_0$ |
| 8 | measure | $q_1$ |

*Read statically. Grey measurements are terminal.*

## ④ QDG




*Shaded nodes are the QDG's backward reach, and the dashed arrow is the classical PDG's last writer.*

## ⑤ Slice of $q_0$

QDG PDG

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
7 circuit.cx(0,1)
8 circuit.measure_all()
```

**QDG** keeps the faulty line 6
**classical PDG** keeps line 5 only

*A full square marks a quantum line, a light one a classical line that builds the circuit.*